



higher education & training

UltraDeep.co.za

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE (VOCATIONAL)
**BASIC PRINCIPLES OF COMPUTER PROGRAMMING
AND COMPUTER LITERACY**
NQF LEVEL 2
(Second paper)

(10041062)

4 March 2026 (Y-paper)
13:00–17:00

This question paper consists of 11 pages.

tvetspapers.co.za **ultradeep.co.za**



Search for NCV Exam Papers

DEPARTMENT OF HIGHER EDUCATION AND TRAINING
REPUBLIC OF SOUTH AFRICA
NATIONAL CERTIFICATE (VOCATIONAL)
BASIC PRINCIPLES OF COMPUTER PROGRAMMING
AND COMPUTER LITERACY
NQF LEVEL 2
(Second paper)
TIME: 4 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer all the questions.
2. Read all the questions carefully.
3. Number the answers according to the numbering system used in this question paper.

INSTRUCTIONS TO STUDENTS AND LECTURERS

1. Students must be seated in the computer laboratory 30 minutes before the start of the examination to ensure that all computers have the appropriate programs.
2. These questions must be completed on a computer by making use of:
 - Scratch
 - Thonny,
 - PyCharm
 - Visual Studio Code
 - or any python IDE

No handwritten code or hand drawn diagrams will be marked!

3. Students must make sure that they print their work immediately after completing a question. Ensure that your EXAMINATION NUMBER appears on each printout. No questions without a printed EXAMINATION NUMBER will be marked.
4. In the event of problems such as a power failure, computer breakdown or a printer breakdown, the invigilator will make the necessary arrangements to continue with the examination process once the problem has been resolved. Students will not be penalized with time lost under these circumstances.
5. All work on the computer should be saved at regular intervals to prevent loss of work. No additional time may be allowed for work lost due to lack of saving.

62)

-3-

6. No student may print his/her work for another student or make a memory stick available to another student or access another student's work on the network. Any attempt to access any information from or transfer information to another student in whatever manner is a contravention of the examination rules and regulations and will be viewed in an extremely serious light.
 7. All printouts or screen prints done during the session must be handed in.
 8. Make sure that your examination number appears on all printouts.
 9. Students may not copy the source code from the python IDE code editor to MS Word and then print it from the word processor.
 10. Steps to print a user interface or any other screenshot
 - 10.1 Run the solution.
 - 10.2 Hold down the ALT key and press the print screen (Print Scrn) key.
 - 10.3 Open a blank MS Word document and paste the screenshot.
 - 10.4 Add your EXAMINATION NUMBER against the left margin.
 - 10.5 Make a printout of the page.
-



QUESTION 1

In a small town, Dr. Mathebula runs a busy clinic that often sees patients with heart issues and fevers. Due to the high volume of patients and limited medical equipment, Dr. Mathebula needs a way to monitor vital signs without relying on physical devices.

To solve this problem, she asks you create a virtual health monitor using Scratch. The goal is to simulate a system that tracks heart rate and temperature, alerting her if a patient's vitals fall outside a healthy range.

Using Scratch, develop a program that prompts the user to enter the heart rate and temperature readings. The program then checks these values against predefined healthy thresholds. If the readings are abnormal, the system displays an alert message, allowing Dr. Mathebula to prioritize patients who need immediate attention.

Use the IPO chart below to assist you in completing the application:

INPUT	PROCESS	OUTPUT
heart rate temperature	Declare all variables Prompt for the heart rate Prompt for the temperature Validate If temperature is greater 37.5 or temperature is less than 37.5 Display:" Your Temperature Is Normal" Else Display:" Your Temperature Is abnormal" If heart rate is greater 60 or heart rate is less than 100 Display:" Your heart rate is Normal" Else Display:" Your heart rate Is abnormal"	Display: results



62)

-5-



Below is a screenshot of what the application should look like:

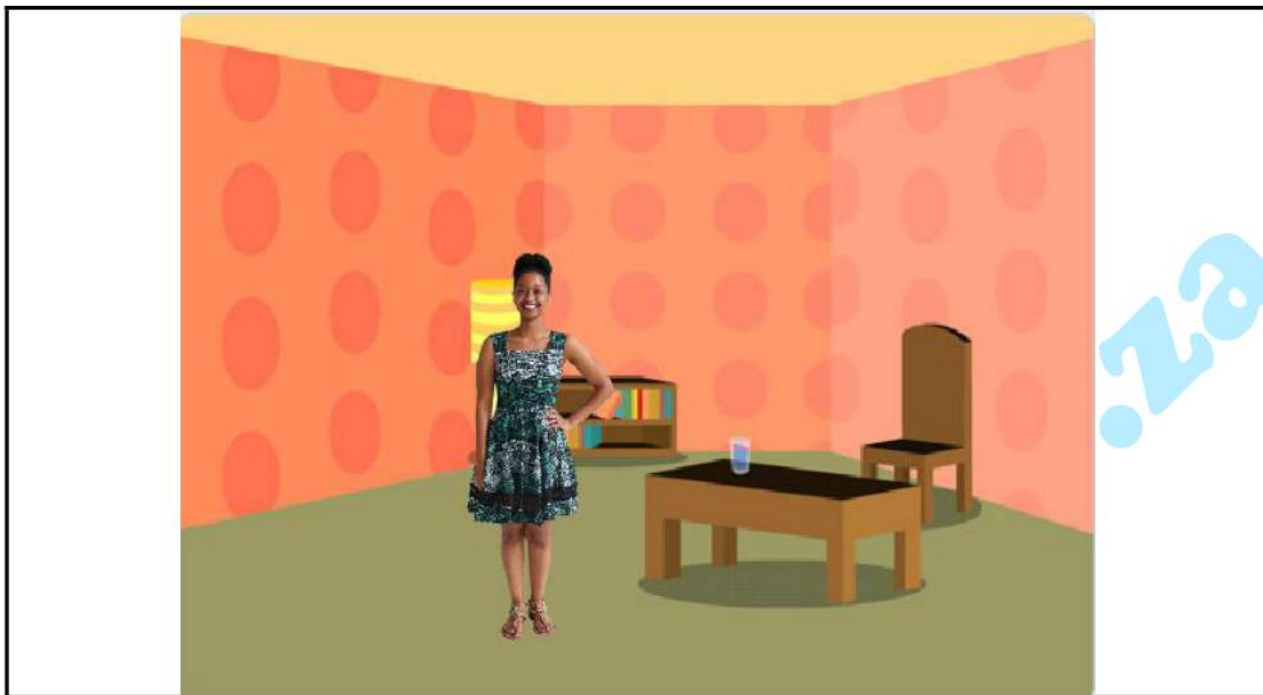


FIGURE 1.1

The following STEPS explain how the application should be used:

STEP 1: Run the application. Enter the heart rate.

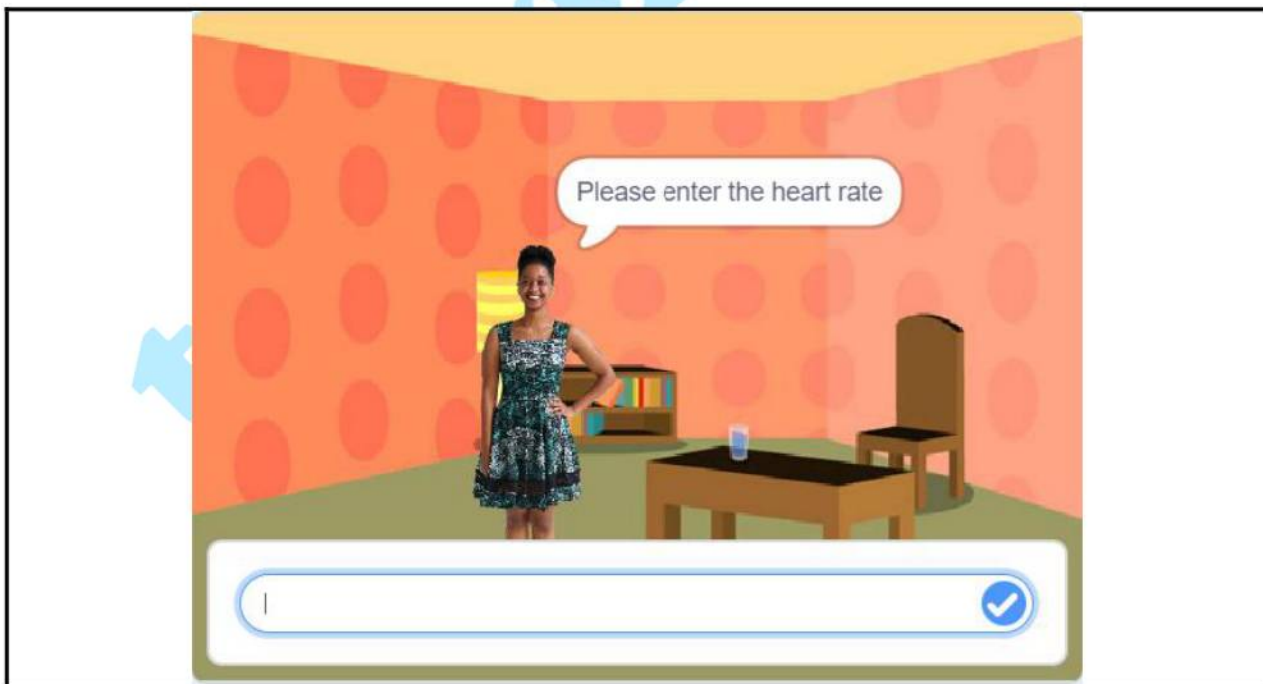


FIGURE 1.2



62)

-6-



STEP 2: Enter the temperature.

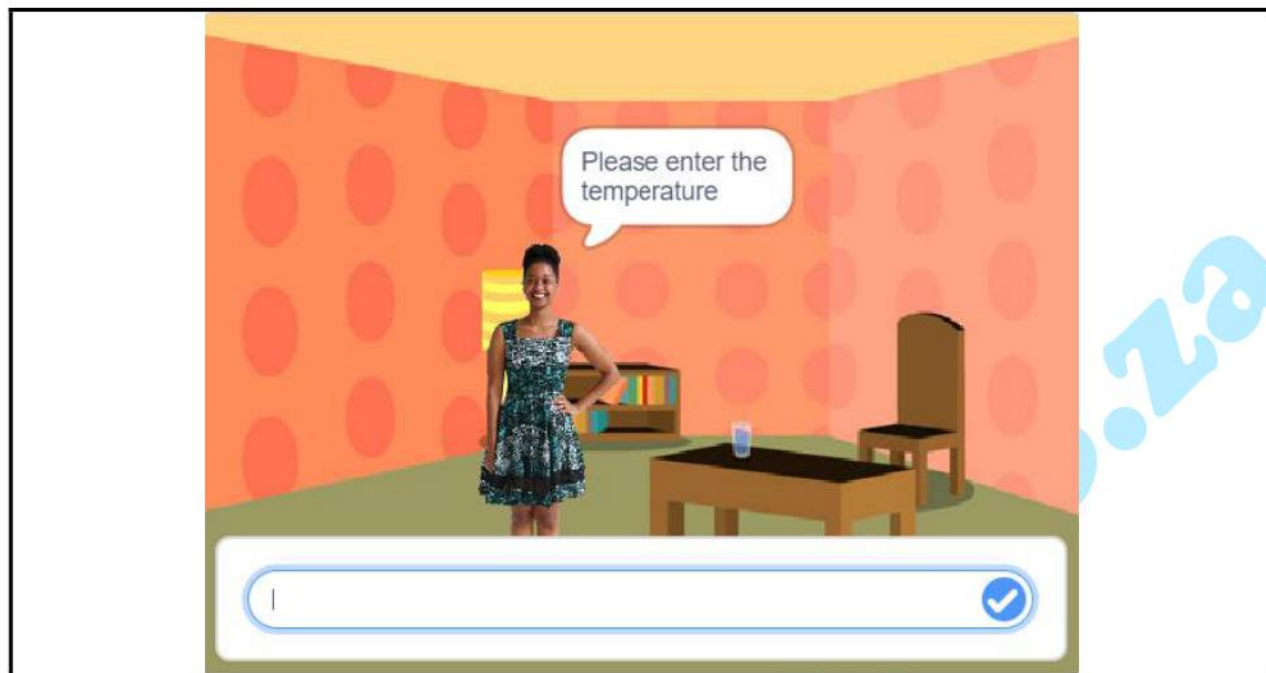


FIGURE 1.3

STEP 3: Determine the heart rate and temperature readings.



FIGURE 1.4



62)

-7-

Follow the instructions below to create the Scratch application:

- 1.1 Set Up the Scratch Environment, Open Scratch and create a new project and name it QUESTION1_EXAMINATION_NUMBER.sb3 (1)
 - 1.2 Choose an appropriate sprite. (2)
 - 1.3 Select an appropriate background. (2)
 - 1.4 Declare all variables and initialise them to 0 (8)
 - 1.5 Prompt the user to enter the heart rate and assign it to its variable. (4)
 - 1.6 Prompt the user to enter the temperature and assign it to its variable. (4)
 - 1.7 Using condition statements, if else if. Determine the whether the heart rate and temperature fall within the threshold and display the results. (13)
 - 1.8 SAVE your program as: QUESTION1_EXAMINATION_NUMBER.sb3 (1)
- Print your program code from Scratch. [35]





QUESTION 2

Study the following Turtle graphics program output. The triangle spiral consists of 10 lines.

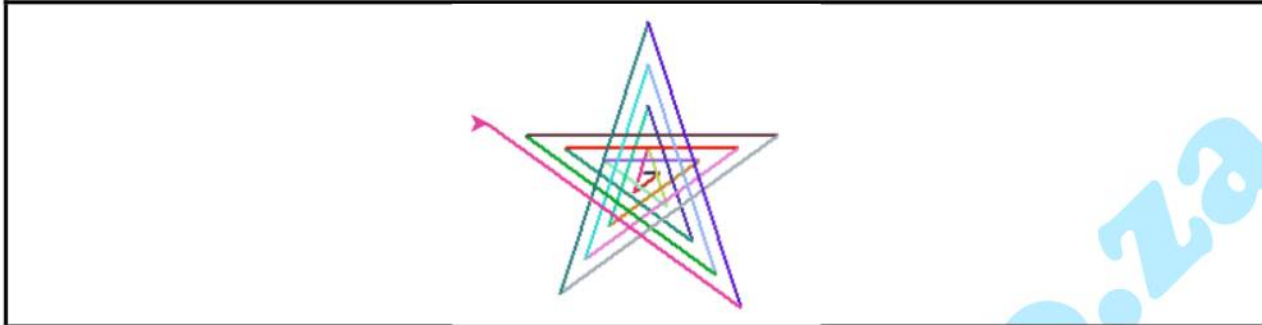


FIGURE 2.1

Write the code in python using the turtle graphics library to draw a shape that will resemble the one above.

- 2.1 Open your preferred python development environment or text editor and create a new project and name it QUESTION2_EXAMINATION_NUMBER.py also add your EXAMINATION NUMBER as a comment on the first line in the code. (1)
- 2.2 The turtle object should be name 'triangle'. (2)
- 2.3 Set the speed for the drawing to 0 (2)
- 2.4 The thickness of the lines must be set to 2 pixels. (2)
- 2.5 The pattern must be repeated of 20 times. (5)
- 2.6 The angle of the drawing must be set at 144 degrees. (5)
- 2.7 The colours used for each line segment must be generated as random colours. (5)

SAVE your program as: QUESTION2_EXAMINATION_NUMBER.py

ADD your examination number as a comment to the first line of the program.

Print your program code from the IDE.

Ensure that the line numbers are printed as well.

[22]





QUESTION 3

The City of Aqua Ville introduces a Sustainable Energy Monitoring System to promote responsible energy usage among businesses. The system calculates monthly energy bills while encouraging the adoption of eco-friendly practices. In response to increasing energy demands, the city has implemented tiered billing rates and rewards for businesses that use energy efficiently.

The program should process multiple businesses in one session and provide clear, structured summaries of their energy usage and costs.

Each business is required to input:

- Business Name (e.g., "GreenTech Ltd.")
- Number of Employees (e.g., 50 employees)
- Energy Usage (kilowatt-hours per month) (e.g., 15,000 kWh)
- Green Energy Adoption Percentage (e.g., 25%)

Base Charge: Every business pays a standard infrastructure maintenance fee of R500 monthly.

- Up to 10,000 kWh: R0.50 per kWh
- 10,001–20,000 kWh: R0.75 per kWh for energy exceeding 10,000 kWh
- Above 20,000 kWh: R1.00 per kWh for energy exceeding 20,000 kWh

Eco-Friendly Rebate: Businesses receive a 5% rebate on their energy costs for every 10% of green energy adoption, capped at 50%.

Refer to the IPO chart below to assist you with your program

INPUT	PROCESS	OUTPUT
- Business Name -Energy Usage (kWh) -Green Energy Adoption (%) Base Charge	Calculate energy cost for each tier Apply green energy rebate Calculate total bill for each business Accumulate total bills for all businesses	Business Name Energy Usage (kWh) Green Energy Rebate Total Energy Bill (Rands)





The following STEPS explain how the application should be used:

STEP 1: Run the application.

The user will be prompted to enter the following information

- Business Name
- Number of Employees
- Energy Usage
- Green Energy Adoption Percentage

```
Enter Business Name: EcoTech Solutions
Enter Number of Employees: 45
Enter Energy Usage (kWh per Month): 12000
Enter Green Energy Adoption Percentage (%): 30
```

FIGURE 3.1

STEP 2: The program will calculate and display the total bill.

```
Billing Summary for EcoTech Solutions:
Number of Employees: 45
Energy Usage: 12000.00 kWh
Base Charge: R500.00
Energy Charge: R6500.00
Green Energy Rebate: R21000.00
Total Bill: R-14000.00
```

FIGURE 3.2

STEP 3: The program will ask if you'd like to enter another business, if yes, you will be prompted to enter the details again. If the user replies no, then the total bill for all businesses will be calculated and displayed.

```
Do you want to enter another business's details? (yes/no): yes
Enter Business Name: Solar Innovations
Enter Number of Employees: 22
Enter Energy Usage (kWh per Month): 9500
Enter Green Energy Adoption Percentage (%): 20

Billing Summary for Solar Innovations:
Number of Employees: 22
Energy Usage: 9500.00 kWh
Base Charge: R500.00
Energy Charge: R4750.00
Green Energy Rebate: R10500.00
Total Bill: R-5250.00

Do you want to enter another business's details? (yes/no):

Total Bill for All Businesses: R-19250.00
Exiting Sustainable Energy Monitoring System...
```



FIGURE 3.3

Follow the instructions below to create the application:



- 3.1 Open your python development environment or text editor.
Create a new python source file Question3_EXAMINATION NUMBER.py (1)
- 3.2 Declare a global variable named total_bill and initialize it to 0. (2)
- 3.3 Create a non-parameterized function named calculate_bill().
Inside this function prompt the user to enter:
 - Business Name
 - Number of Employees
 - Energy Usage
 - Green Energy Adoption Percentage
 (6)
- 3.4 Perform validation to check if the energy usage is less or equal to 10000 kWh.
the water charge is R0.50 per kWh. If the energy usage is between 10001 and
20000 kWh, then the energy charge is R0.75 per kWh. Then if the energy usage
is greater than 20000 kWh, the energy charge is R1.00 per kWh. (8)
- 3.5 Implement a condition for businesses to receive a 5% rebate on their energy
costs for every 10% of green energy adoption, capped at 50%. (4)
- 3.6 Calculate the total bill using the inputs from the user. (4)
- 3.7 Implement a loop that allows the user to enter multiple businesses. (7)
- 3.8 Calculate the total bill for all businesses is accumulated in total_bill. (4)
- 3.9 Call the function calculate_bill() to perform and display energy bill Calculations (4)
- 3.10 Test your program by entering the data on FIGURE 3.1 and FIGURE 3.3 (2)
- 3.11 SAVE your program as: Question3_EXAMINATION_NUMBER.py (1)

ADD your examination number as a comment to the first line of the program.

Print your program code from the IDE.

Ensure that the line numbers are printed as well.

[43]

TOTAL: 100

